

WEEK 1

LINUX BASICS AND SYSTEM STARTUP

Linux Fundamentals Course Documentation

Overview

This repository contains complete course materials for learning Linux fundamentals, system operations, graphical interfaces, command-line usage, file management, and software installation.

The course is designed for:

- * Beginners learning Linux for the first time*
- * Students pursuing IT and computer science studies*
- * System administrators and IT professionals*
- * Anyone interested in open-source operating systems*

The course combines theory, practical examples, and hands-on laboratory exercises to build strong Linux administration skills.

Course Learning Outcomes

By completing this course, learners will be able to:

- * Understand Linux architecture and philosophy*
- * Work efficiently with the Linux command line*
- * Navigate and manage the Linux filesystem*
- * Configure Linux systems using graphical tools*

- * Install and manage software packages*
- * Perform basic Linux system administration tasks*
- * Manage files, directories, and permissions*
- * Use Linux utilities and command-line tools effectively*

Course Structure

01. Course Introduction

About This Course

This course introduces learners to the Linux operating system and its practical usage in modern computing environments. It covers both graphical and command-line interfaces while focusing on real-world system administration tasks.

Topics Covered

- * Introduction to Linux*
- * Linux distributions*
- * Course objectives*
- * Linux use cases*
- * Open-source ecosystem*

02. Linux Philosophy and Concepts

Core Concepts

- * Open Source Software*

- * *Linux Architecture*
- * *Kernel and Shell*
- * *Multiuser Environment*
- * *Multitasking*
- * *Linux Filesystem Hierarchy*

Philosophy and Principles

- * *Everything is a File*
- * *Small Modular Tools*
- * *Command-Line Efficiency*
- * *User Permissions and Security*
- * *Flexibility and Portability*

03. Linux Basics and System Startup

System Startup and Essentials

- * *Linux Boot Process*
- * *BIOS and UEFI*
- * *Bootloader Basics*
- * *System Initialization*
- * *User Login Sessions*

Startup and Service Management

- * *Runlevels and Targets*
- * *System Services*
- * *Daemons*

- * Reboot and Shutdown Procedures*

- * Startup Troubleshooting*

04. Graphical Interface

The Desktop Experience

- * Linux Desktop Environments*

- * File Browsing*

- * Windows and Panels*

- * Workspaces*

- * Desktop Personalization*

Desktop Environment Management

- * GUI Utilities*

- * Theme and Appearance Settings*

- * Accessibility Options*

- * Workspace Management*

- * Display Configuration*

05. System Configuration from the Graphical Interface

GUI-Based System Management

- * User Management*

- * Display Settings*

- * Network Configuration*

- * Time and Region Settings*

- * Device and Printer Setup*

Additional Configuration Tasks

- * Software Management*

- * Updates and Package Tools*

- * Startup Applications*

- * System Monitoring*

- * Security Settings*

06. Common Applications

Essential Linux Applications

- * Web Browsers*

- * Office Applications*

- * Media Players*

- * Compression Utilities*

- * Screenshot Tools*

Productivity and Development Tools

- * Text Editors*

- * IDEs and Development Tools*

- * Terminal Applications*

- * Communication Tools*

- * Backup Utilities*

07. Command Line Operations

Chapter Overview

This chapter introduces Linux command-line operations and teaches learners how to manage files, directories, applications, and software packages efficiently using terminal commands.

Learning Objectives

After completing this chapter, learners will be able to:

- * Use terminal commands effectively*
- * Navigate the filesystem*
- * Manage files and directories*
- * Search for files*
- * Redirect input and output*
- * Install software packages*

Command-Line Mode Options

Topics Covered

- * Introduction to the Command Line*
- * Using a Text Terminal on the Graphical Desktop*
- * Launching Terminal Windows*

- * Basic Linux Utilities*
- * The Linux Command Prompt*
- * Using sudo*
- * Configuring sudo Access*
- * Switching Between GUI and CLI*
- * Virtual Terminals*
- * Disabling the Graphical Interface*

Lab 7.1

Killing the Graphical User Interface

Basic Operations

Topics Covered

- * Logging In and Out*
- * Rebooting and Shutting Down*
- * Locating Applications*
- * Accessing Directories*
- * Absolute and Relative Paths*
- * Exploring the Filesystem*
- * Hard Links*
- * Symbolic Links*
- * Directory Navigation History*

Practice Exercises

- * Accessing Directories Using Command Prompt*

Lab 7.2

*****Locating Applications*****

Working with Files

Topics Covered

** Viewing Files*

** File Manipulation Commands*

** `touch` Command*

** `mkdir` and `rmdir`*

** Moving and Renaming Files*

** Removing Files and Directories*

** Modifying the Shell Prompt*

** File and Directory Operations*

Practice Exercises

** Working With Files and Directories Using the Command Prompt*

Lab 7.3

*****Creating, Moving and Removing Files*****

Searching for Files

Topics Covered

- * Standard File Streams*
- * Input and Output Redirection*
- * Pipes*
- * `locate` Command*
- * Wildcards*
- * Filename Matching*
- * `find` Command*
- * Advanced `find` Options*
- * Searching by Time and Size*

Practice Exercises

- * Locating Files*
- * Using Wildcards with `ls`*

Lab 7.4

Finding Directories and Creating Symbolic Links

Installing Software

Topics Covered

- * Linux Package Management Systems*
- * Low-Level Package Management*
- * `dpkg` Package Manager*

* `rpm` Package Manager
* `dnf` Package Manager
* `zypper` Package Manager
* `apt` Package Manager

Lab 7.5

****Installing and Removing Software Packages****

Practical Labs

Lab No	Lab Title	
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Lab 7.1	Killing the Graphical User Interface	
Lab 7.2	Locating Applications	
Lab 7.3	Creating, Moving and Removing Files	
Lab 7.4	Finding Directories and Creating Symbolic Links	
Lab 7.5	Installing and Removing Software Packages	

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